



National Load Despatch Centre
राष्ट्रीय भार प्रेषण केंद्र
GRID CONTROLLER OF INDIA LIMITED
ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड

(Government of India Enterprise/ भारत सरकार का उद्यम)
B-9, QUTUB INSTITUTIONAL AREA, KATWARIA SARAI, NEW DELHI -110016
बी-9, कुतुब इंस्टीट्यूशनल एरिया, कटवारिया सराये, न्यू दिल्ली-110016

Ref: POSOCO/NLDC/SO/Daily PSP Report

दिनांक: 21th April 2024

To,

- कार्यकारी निदेशक, पू.क्षे.भा.प्रे.के., 14, गोल्फ क्लब रोड, कोलकाता - 700033
Executive Director, ERLDC, 14 Golf Club Road, Tollygunge, Kolkata, 700033
- कार्यकारी निदेशक, ऊ.क्षे.भा.प्रे.के., 18/ए, शाहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली - 110016
Executive Director, NRLDC, 18-A, Shaheed Jeet Singh Marg, Katwaria Sarai, New Delhi - 110016
- कार्यकारी निदेशक, प.क्षे.भा.प्रे.के., एफ3, एम आई डी सी क्षेत्र, अंधेरी, मुंबई -400093
Executive Director, WRLDC, F-3, M.I.D.C. Area, Marol, Andheri (East), Mumbai-400093
- कार्यकारी निदेशक, ऊ.पू.क्षे.भा.प्रे.के., डोंगतेह, लोअर नोंग्रह, लापालंग, शिलोंग - 793006
Executive Director, NERLDC, Dongteih, Lower Nongrah, Lapalang, Shillong - 793006, Meghalaya
- कार्यकारी निदेशक, द.क्षे.भा.प्रे.के., 29, रेस कोर्स क्रॉस रोड, बंगलुरु -560009
Executive Director, SRLDC, 29, Race Course Cross Road, Bangalore-560009

Sub: Daily PSP Report for the date 20.04.2024.

महोदय/Dear Sir,

आई०ई०जी०सी०-2010 की धारा स.-5.5.1 के प्रावधान के अनुसार, दिनांक 20-अप्रैल-2024 की अखिल भारतीय प्रणाली की दैनिक ग्रिड निष्पादन रिपोर्ट रा०भा०प्रे०के० की वेबसाइट पर उपलब्ध है।

As per article 5.5.1 of the Indian Electricity Grid Code, the daily report pertaining power supply position of All India Power System for the date 20th April 2024, is available at the NLDC website.

धन्यवाद,

Report for previous day

Date of Reporting: 21-Apr-2024

A. Power Supply Position at All India and Regional level

	NR	WR	SR	ER	NER	TOTAL
Demand Met during Evening Peak hrs(MW) (at 20:00 hrs; from RLDCs)	59825	62914	51661	28374	2539	205313
Peak Shortage (MW)	0	0	0	0	0	0
Energy Met (MU)	1284	1596	1279	637	48	4844
Hydro Gen (MU)	173	40	38	29	11	291
Wind Gen (MU)	16	115	58	-	-	189
Solar Gen (MU)*	160.72	91.61	106.55	3.59	0.77	363
Energy Shortage (MU)	0.00	0.00	0.00	2.94	0.00	2.94
Maximum Demand Met During the Day (MW) (From NLDC SCADA)	61026	71514	60434	29818	2739	212500
Time Of Maximum Demand Met	19:47	15:21	14:47	22:36	18:43	22:36

B. Frequency Profile (%)

Region	FVI	< 49.7	49.7 - 49.8	49.8 - 49.9	< 49.9	49.9 - 50.05	> 50.05
All India	0.043	0.14	0.78	4.94	5.86	75.26	18.89

C. Power Supply Position in States

Region	States	Max.Demand Met during the day (MW)	Shortage during maximum Demand (MW)	Energy Met (MU)	Drawal Schedule (MU)	OD(+)/UD(-) (MU)	Max OD (MW)	Energy Shortage (MU)
NR	Punjab	7356	0	151.4	54.8	-1.4	142	0.00
	Haryana	8719	0	164.9	97.0	-2.4	292	0.00
	Rajasthan	13396	0	276.9	69.8	0.0	508	0.00
	Delhi	4640	0	100.5	92.2	0.7	239	0.00
	UP	23098	0	458.5	198.8	-2.4	470	0.00
	Uttarakhand	2191	0	43.5	23.2	-0.6	86	0.00
	HP	1565	0	31.0	14.0	-0.3	40	0.00
	J&K(UT) & Ladakh(UT)	2377	0	48.7	34.9	0.1	179	0.00
	Chandigarh	228	0	4.5	4.7	-0.3	7	0.00
WR	Railways_NR ISTS	188	0	4.0	3.8	0.1	64	0.00
	Chhattisgarh	6169	0	138.4	77.6	-1.0	269	0.00
	Gujarat	22312	0	474.3	173.8	-3.6	618	0.00
	MP	12623	0	279.1	138.1	-5.4	427	0.00
	Maharashtra	27734	0	629.3	224.9	1.6	671	0.00
	Goa	740	0	14.8	10.9	3.6	79	0.00
	DNHDDPDCL	1238	0	28.9	29.3	-0.4	35	0.00
	AMNSIL	875	0	18.7	5.7	0.2	279	0.00
SR	BALCO	522	0	12.4	12.5	-0.1	85	0.00
	Andhra Pradesh	13111	0	249.9	91.6	-1.8	481	0.00
	Telangana	10426	0	221.4	109.1	-1.9	769	0.00
	Karnataka	13641	0	276.0	142.8	-1.1	1017	0.00
	Kerala	5288	0	105.9	87.5	1.0	358	0.00
	Tamil Nadu	19537	0	415.3	269.5	-5.6	359	0.00
	Puducherry	488	0	10.6	10.2	-0.2	32	0.00
ER	Bihar	6703	0	128.8	118.4	-0.9	458	1.46
	DVC	3233	0	73.9	-44.1	0.1	395	0.00
	Jharkhand	2079	0	41.2	34.1	-2.2	357	1.49
	Odisha	6372	0	140.0	52.8	-1.3	331	0.00
	West Bengal	12056	0	251.8	113.8	-1.2	181	0.00
	Sikkim	82	0	1.4	1.5	-0.1	25	0.00
NER	Railways_ER ISTS	13	0	0.1	0.1	0.0	2	0.00
	Arunachal Pradesh	151	0	2.7	2.5	0.1	41	0.00
	Assam	1628	0	28.1	21.9	0.0	149	0.00
	Manipur	205	0	2.8	2.9	-0.1	20	0.00
	Meghalaya	336	0	5.7	4.3	-0.2	52	0.00
	Mizoram	117	0	1.8	1.7	-0.2	30	0.00
	Nagaland	128	0	2.0	2.2	-0.2	19	0.00
	Tripura	291	0	4.6	5.0	-0.2	77	0.00

D. Transnational Exchanges (MU) - Import(+ve)/Export(-ve)

	Bhutan	Nepal	Bangladesh	Godda -> Bangladesh
Actual (MU)	1.0	-5.9	-24.4	-27.1
Day Peak (MW)	237.0	-411.7	-1066.0	-1436.8

E. Import/Export by Regions (in MU) - Import(+ve)/Export(-ve); OD(+)/UD(-)

	NR	WR	SR	ER	NER	TOTAL
Schedule(MU)	145.5	-349.1	246.6	-46.2	3.2	0.0
Actual(MU)	141.6	-366.0	254.7	-42.0	3.7	-8.0
O/D/U/D(MU)	-3.9	-16.9	8.1	4.3	0.5	-8.0

F. Generation Outage(MW)

	NR	WR	SR	ER	NER	TOTAL	% Share
Central Sector	3757	5321	4418	3426	155	17076	52
State Sector	6066	6022	2368	1404	180	16039	48
Total	9823	11342	6786	4829	335	33115	100

G. Sourcewise generation (Gross) (MU)

	NR	WR	SR	ER	NER	All India	% Share
Coal	793	1685	737	741	15	3971	75
Lignite	22	17	68	0	0	108	2
Hydro	173	40	38	29	11	291	6
Nuclear	25	64	76	0	0	165	3
Gas, Naptha & Diesel	32	80	6	0	25	143	3
RES (Wind, Solar, Biomass & Others)	189	211	184	5	1	589	11
Total	1234	2097	1109	776	51	5266	100

Share of RES in total generation (%)	15.28	10.04	16.59	0.68	1.55	11.19
Share of Non-fossil fuel (Hydro,Nuclear and RES) in total generation(%)	31.33	15.02	26.82	4.45	22.42	19.84

H. All India Demand Diversity Factor

Based on Regional Max Demands	1.061
Based on State Max Demands	1.091

I. All India Peak Demand and shortage at Solar and Non-Solar Hour

	Max Demand Met(MW)	Time	Shortage(MW)
Solar hr	209256	15:21	0
Non-Solar hr	212500	22:36	0

Diversity factor = Sum of regional or state maximum demands / All India maximum demand

**Note: All generation MU figures are gross

***Godda (Jharkhand) -> Bangladesh power exchange is through the radial connection (isolated from Indian Grid)

Solar Hours -> 07:00 to 17:00hrs and rest are Non-Solar Hours

*Source: RLDCs for solar connected to ISTS; SLDCs for embedded solar. Limited visibility of embedded solar data.

Executive Director-NLDC

INTER-REGIONAL EXCHANGES

Import=(+ve) /Export =(-ve) for NET (MU)

Date of Reporting: 21-Apr-2024

Sl No	Voltage Level	Line Details	No. of Circuit	Max Import (MW)	Max Export (MW)	Import (MU)	Export (MU)	NET (MU)
Import/Export of ER (With NR)								
1	HVDC	ALIPURDUAR-AGRA	2	0	0	0.0	0.0	0.0
2	HVDC	PUSAULI B/B	-	0	48	0.0	1.3	-1.3
3	765 kV	GAYA-VARANASI	2	828	526	0.8	0.0	0.8
4	765 kV	SASARAM-FATEHPUR	1	195	368	0.0	3.7	-3.7
5	765 kV	GAYA-BALIA	1	67	539	0.0	7.5	-7.5
6	400 kV	PUSAULI-VARANASI	1	0	95	0.0	1.0	-1.0
7	400 kV	PUSAULI -ALLAHABAD	1	50	38	0.0	0.1	-0.1
8	400 kV	MUZAFFARPUR-GORAKHPUR	2	668	386	1.8	0.0	1.8
9	400 kV	PATNA-BALIA	2	361	800	0.0	8.7	-8.7
10	400 kV	NAUBATPUR-BALIA	2	0	246	0.0	2.4	-2.4
11	400 kV	BIHARSHARIFF-BALIA	2	514	92	4.7	0.0	4.7
12	400 kV	MOTIHARI-GORAKHPUR	2	309	422	0.0	3.5	-3.5
13	400 kV	BIHARSHARIFF-VARANASI	2	221	139	0.6	0.0	0.6
14	220 kV	SAHUPURI-KARAMNASA	1	73	90	0.0	0.7	-0.7
15	132 kV	NAGAR UNTARI-RIHAND	1	0	0	0.0	0.0	0.0
16	132 kV	GARWAH-RIHAND	1	30	0	0.8	0.0	0.8
17	132 kV	KARMANASA-SAHUPURI	1	0	0	0.0	0.0	0.0
18	132 kV	KARMANASA-CHANDAULI	1	0	0	0.0	0.0	0.0
ER-NR						8.6	28.9	-20.3
Import/Export of ER (With WR)								
1	765 kV	JHARSUGUDA-DHARAMJAIGARH	4	1892	0	30.9	0.0	30.9
2	765 kV	NEW RANCHI-DHARAMJAIGARH	2	1193	0	17.2	0.0	17.2
3	765 kV	JHARSUGUDA-DURG	2	0	715	0.0	11.2	-11.2
4	400 kV	JHARSUGUDA-RAIGARH	4	0	406	0.0	5.7	-5.7
5	400 kV	RANCHI-SIPAT	2	225	68	2.4	0.0	2.4
6	220 kV	BUDHIPADAR-RAIGARH	1	46	96	0.0	0.7	-0.7
7	220 kV	BUDHIPADAR-KORBA	2	0	0	0.0	0.0	0.0
ER-WR						50.4	17.5	32.9
Import/Export of ER (With SR)								
1	HVDC	JEYPORE-GAZUWAKA B/B	2	0	590	0.0	13.7	-13.7
2	HVDC	TALCHER-KOLAR BIPOLE	2	0	1490	0.0	36.1	-36.1
3	765 kV	ANGUL-SRIKAKULAM	2	0	2879	0.0	57.7	-57.7
4	400 kV	TALCHER-I/C	2	424	256	8.2	0.0	8.2
5	220 kV	BALIMELA-UPPER-SILERRU	1	0	0	0.0	0.0	0.0
ER-SR						0.0	107.5	-107.5
Import/Export of ER (With NER)								
1	400 kV	BINAGURI-BONGAIGAON	2	427	135	3.0	0.5	2.5
2	400 kV	ALIPURDUAR-BONGAIGAON	2	731	311	4.9	0.0	4.9
3	220 kV	ALIPURDUAR-SALAKATI	2	113	56	0.8	0.0	0.8
ER-NER						8.7	0.5	8.3
Import/Export of NER (With NR)								
1	HVDC	BISWANATH CHARIALI-AGRA	2	656	0	11.4	0.0	11.4
NER-NR						11.4	0.0	11.4
Import/Export of WR (With NR)								
1	HVDC	CHAMPA-KURUKSHETRA	2	0	2010	0.0	48.2	-48.2
2	HVDC	VINDHYACHAL B/B	-	488	0	12.2	0.0	12.2
3	HVDC	MUNDRA-MOHINDERGARH	2	0	1450	0.0	29.8	-29.8
4	765 kV	GWALIOR-AGRA	2	122	2042	0.0	25.2	-25.1
5	765 kV	GWALIOR-PHAGI	2	1115	1242	5.4	11.1	-5.7
6	765 kV	JABALPUR-ORAI	2	347	792	0.0	16.4	-16.4
7	765 kV	GWALIOR-ORAI	1	537	0	7.7	0.0	7.7
8	765 kV	SATNA-ORAI	1	0	1059	0.0	20.0	-20.0
9	765 kV	BANASKANTHA-CHITORGARH	2	1750	469	11.4	1.9	9.5
10	765 kV	VINDHYACHAL-VARANASI	2	0	3017	0.0	53.2	-53.2
11	400 kV	ZERDA-KANKROLI	1	368	34	3.2	0.1	3.2
12	400 kV	ZERDA -BHINMAL	1	367	33	3.2	0.1	3.2
13	400 kV	VINDHYACHAL -RIHAND	1	990	0	22.1	0.0	22.1
14	400 kV	RAPP-SHUJALPUR	2	661	297	4.2	2.2	2.1
15	400 kV	NEEMUCH-Chittorgarh	2	546	216	3.7	1.3	2.4
16	220 kV	BHANPURA-RANPUR	1	0	126	0.0	2.4	-2.4
17	220 kV	BHANPURA-MORAK	1	0	30	0.0	2.1	-2.1
18	220 kV	MEHGAON-AURAIYA	1	83	0	1.2	0.0	1.2
19	220 kV	MALANPUR-AURAIYA	1	64	0	0.7	0.0	0.7
20	132 kV	GWALIOR-SAWAI MADHOPUR	1	0	0	0.0	0.0	0.0
21	132 kV	RAJGHAT-LALITPUR	2	0	0	0.0	0.0	0.0
WR-NR						75.1	213.7	-138.6
Import/Export of WR (With SR)								
1	HVDC	BHADRAWATI B/B	-	0	1006	0.0	11.7	-11.7
2	HVDC	RAIGARH-PUGALUR	2	0	5512	0.0	69.6	-69.6
3	765 kV	SOLAPUR-RAICHUR	2	0	1883	0.0	28.8	-28.8
4	765 kV	WARDHA-NIZAMABAD	2	0	2409	0.0	45.1	-45.1
5	765 kV	WARORA-WARANGAL(NEW)	2	0	2857	0.0	54.8	-54.8
6	400 kV	KOLHAPUR-KUDGI	2	895	0	11.6	0.0	11.6
7	220 kV	KOLHAPUR-CHIKODI	2	0	0	0.0	0.0	0.0
8	220 kV	PONDA-AMBEWADI	1	0	97	2.3	0.0	2.3
9	220 kV	XELDEM-AMBEWADI	1	0	120	2.2	0.0	2.2
WR-SR						16.1	210.0	-193.9

INTERNATIONAL EXCHANGES					Import(+ve)/Export(-ve)	
State	Region	Line Name	Max (MW)	Min (MW)	Avg (MW)	Energy Exchange (MU)
BHUTAN	ER	400kV MANGDECHHU-ALIPURDUAR 1,2&3 i.e. ALIPURDUAR RECEIPT (from MANGDECHU HEP 4*180MW)	278	-31	81	1.94
	ER	400kV TALA-BINAGURI 1,2,4 (& 400kV MALBASE - BINAGURI) i.e. BINAGURI RECEIPT (from TALA HEP 6*170MW)	154	-85	65	1.55
	ER	220kV CHUKHA-BIRPARA 1&2 (& 220kV MALBASE - BIRPARA) i.e. BIRPARA RECEIPT (from CHUKHA HEP 4*84MW)	-172	-79	-130	-3.11
	NER	132kV GELEPHU-SALAKATI	-13	7	-2	-0.04
	NER	132kV MOTANGA-RANGIA	40	0	28	0.67
NEPAL	NR	NEPAL IMPORT (FROM UP)	9	0	39	0.94
	NR	132kV MAHENDRANAGAR-TANAKPUR(NHPC)	-78	0	-39	-0.93
	ER	NEPAL IMPORT (FROM BIHAR)	39	0	-11	-0.25
	ER	400kV DHALKEBAR-MUZAFFARPUR 1&2	-460	0	-238	-5.70
BANGLADESH	ER	BHERAMARA B/B HVDC (B'DESH)	-946	-791	-909	-21.81
	ER (Isolated from Indian Grid)	400kV GODDA_TPS-RAHANPUR (B'DESH) D/C	-1437	-814	-1128	-27.06
	NER	132kV COMILLA-SURAJMANI NAGAR 1&2	-120	0	-108	-2.58

CROSS BORDER EXCHANGE SCHEDULE

Date of Reporting: 21-Apr-2024

Export From India (in MU)

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Nepal	0.35	0.00	6.37	0.00	0.00	0.00	0.00	0.00	6.72
Bangladesh	21.76	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.76
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Export	22.11	0.00	6.37	0.00	0.00	0.00	0.00	0.00	28.48

Import by India(in MU)

Country	GNA (ISGA/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.52	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.52
Nepal	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Bangladesh	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Import	0.52	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.52

Net from India(in MU)

-ve : Export / +ve : Import

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.52	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.52
Nepal	-0.35	0.00	-6.37	0.00	0.00	0.00	0.00	0.00	-6.72
Bangladesh	-21.76	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-21.76
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Net	-21.59	0.00	-6.37	0.00	0.00	0.00	0.00	0.00	-27.96